

Department of Computer Science and Engineering
 Faculty of Engineering & Technology, Gurukula Kangri Deemed to be University, Haridwar
 First Sessional Exam
 Subject: Design and Analysis of Algorithm
 Paper Code: BCE-C513

Time: 1 Hr

MM: 20

Note: Answer any three questions from Section A (each question carries 4 marks).

Answer any one question from Section B (each question carries 8 marks).

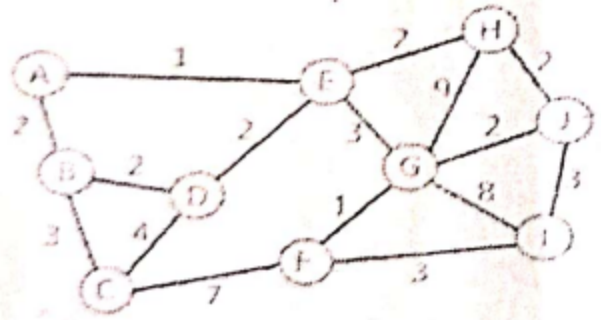
Q.No.	Section A (Questions of 4 marks each)	Keyword	CO	BT
Q1.	Define algorithm? What are its characteristics?	Remembering	(CO1)	BT1
Q2.	Write an algorithm to input the integer values for 3 variables and print their SUM and AVERAGE.	Applying	(CO4)	BT3
Q3.	Explain time and space complexity with the help of example.	Understanding	(CO2)	BT2
Q4.	Explain asymptotic notation (Big Oh, Big Theta, and Big Omega).	Analyzing	(CO6)	BT4
	Section B (Questions of 8 marks each)			
Q1.	Find the time complexity of binary search.	Applying	(CO3)	BT3
Q2.	Solve any two of the following recurrences relation by using Master Method a. $T(n) = T(n/2) + 1$ b. $T(n) = 2T(n/2) + n$ c. $T(n) = 2T(n/2 + 17) + n$	Evaluating	(CO4)	BT5

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Q.No.	Section A (Questions of 4 marks each)	Keyword	CO	BT										
Q1.	Define Dijkstra's Algorithm with suitable example.	Remembering	(CO1)	BT1										
Q2.	Explain Subset Sum Problem in details.	Applying	(CO4)	BT3										
Q3.	Differentiate between Greedy and Dynamic Programming.	Understanding	(CO2)	BT2										
Q4.	Write short note any one of the following: (a). Backtracking (b). MST	Analyzing	(CO6)	BT4										
	Section B (Questions of 8 marks each)													
Q1.	Find the minimum spanning tree using with the prim's algorithm. 	Applying	(CO3)	BT3										
Q2.	0-1 knapsack problem has four item and knapsack capacity is 11. Weight and profit of each item is given in below table. Find the maximum profit using greedy strategy. <table border="1" data-bbox="504 2077 1176 2166"> <tr> <td>P_i</td><td>50</td><td>30</td><td>32</td><td>27</td></tr> <tr> <td>W_i</td><td>5</td><td>6</td><td>4</td><td>3</td></tr> </table>	P_i	50	30	32	27	W_i	5	6	4	3	Evaluating	(CO4)	BT5
P_i	50	30	32	27										
W_i	5	6	4	3										

Attempt any two questions from the following:-

1. Explain Matrix Multiplication using parallel algorithm with the help of example.

2. Write in detail about Hamiltonian cycle. Give example for it.

3. Solve Recurrence Relation using Master Method.
i) $n^2 + 2n + 4$.
ii) $n^3 + 2n^2 + 3n + 3$.